

Readiness to Train

Project Results

KU Leuven & OH Leuven
April 2026

1. Introduction

The goal of this project is to develop a **causal framework for optimising training load per player** in professional football, such that each player arrives at match day in the best possible physical condition. The central question is: *"Given everything we know about a player this morning, what training intensity should we prescribe today to maximise their match-day performance while respecting their biological constraints?"*

This is fundamentally a **causal** question: coaches already make training decisions based on player state, which means the observed data confounds the effect of training with the selection of who receives it. Standard predictive modelling cannot disentangle these; causal methods (G-computation, IPTW, DTR optimisation) are required.

The project uses **daily monitoring data** collected by OH Leuven over approximately 20 months (July 2024 – February 2026), covering **28 first-team players** and totalling **14,359 player-day observations**. Four raw datasets were merged into a single analysis-ready file (RTT.xlsx) containing 46 engineered features per player-day.

Three experiments were conducted, each building on the previous. This report describes the data, the analysis process, the key engineered variables, and the results of all three experiments.

2. Data and Key Variables

Data sources

Dataset	Rows	Scope	Content
Readiness_Data	14,359	28 players, daily	Wellness z-scores, ACWR, GPS benchmarks, medical status
Raw_Data	9,968	84 players, session-level	Detailed GPS, heart rate, session metadata
Sessions	1,206	Team-level	Match day flags, session types
Games	403	24 players, match-level	High-intensity distance, efforts per ball-in-play, minutes

Data analysis and exploration

Before any modelling, we conducted thorough data analysis across three levels of Jupyter notebooks. All visualisations are preserved inside the notebooks and can be re-examined interactively.

Level 0 — Data Quality Checks:

- **0. Processed_Data_Quality** — automated validation of the processed RTT.xlsx dataset: player coverage, column completeness, temporal integrity, ACWR flag verification, and encoding checks.
- **0. TI_Missingness_Analysis** — investigation of why Training Intensity Yesterday has missing values for certain rows: free-day fill logic validation, NaN patterns by activity type.

Level 1 — Data Visualisation & EDA:

- **1.1. Match Analysis** — match-level data exploration from Games.xlsx: match intensity distributions, per-player performance profiles, playing time patterns, and relationships between match load and subsequent wellness responses.
- **1.2. Raw Data Visualisation** — comprehensive EDA across all four raw datasets: missingness heatmaps, dataset linkage (Venn diagrams showing player overlap), temporal coverage, wellness and GPS distributions, per-player radar charts.
- **1.3. Processed Data Visualisation** — EDA of the merged RTT.xlsx: variable distributions across the full season, per-player wellness trajectories, ACWR time series, feature correlation heatmaps.

Level 2 — Experiments (described in Sections 3–5):

- **2.1. Experiment 1** — Match Intensity prediction (reference / exploratory)
- **2.2. Experiment 2** — Training Intensity prediction (primary experiment)
- **2.3. Experiment 3** — Status Decrease prediction (auxiliary experiment)

Key engineered variables

Training Intensity Yesterday (the treatment variable): A composite GPS score capturing how hard a player trained on the previous day. Computed as $\tanh(\text{harmonic_mean}(\text{TD}\%, \text{HSD}\%, \text{Dec}\%, \text{Sprints}\%) / 100)$, where each GPS metric is expressed as a percentage of the player's personal match benchmark. The harmonic mean penalises sessions where one component is disproportionately low (e.g., high distance but no sprints), ensuring the score reflects balanced high-intensity effort. The tanh function soft-caps the score in $[0, 1]$: 0 = full rest, values near 1 = match-equivalent effort. Free days receive exactly 0. The per-player distribution of Training Intensity is shown in the figure below.

Match Intensity Yesterday (the match-day outcome): A composite match performance score computed as the *geometric mean* of two key metrics — high-intensity distance per ball-in-play minute (HID/BIP) and high-intensity efforts per ball-in-play minute (HIE/BIP) — scaled by minutes played: $\sqrt{\text{HID/BIP} \times \text{HIE/BIP}} \times \sqrt{\text{clip}(\text{minutes}, 15, 90) / 90}$. The geometric mean ensures that both distance and effort count equally (a player who runs far but makes few explosive efforts does not score as highly as one who does both). The minutes scaling rewards full-match participation. The distribution of Match Intensity across all matches is shown below. Only filled on the day after a match.

Status Decrease (the short-term outcome): A binary indicator (0 or 1) capturing whether a player's medical status worsened from one day to the next (e.g., Available → Attention, or Attention → Injured). Prevalence is low: approximately 3–5% of observations.

Why these variables? Training Intensity serves as a proxy for the coaching decision (what load was prescribed), while Match Intensity and Status Decrease serve as outcomes (how the player responded). The causal question is whether we can identify the effect of training intensity on player outcomes while accounting for the fact that coaches assign training based on player state (confounding by indication).

Figure 1: Distribution of Match Intensity across all player-match observations. Mean = 3.41, median = 3.34.

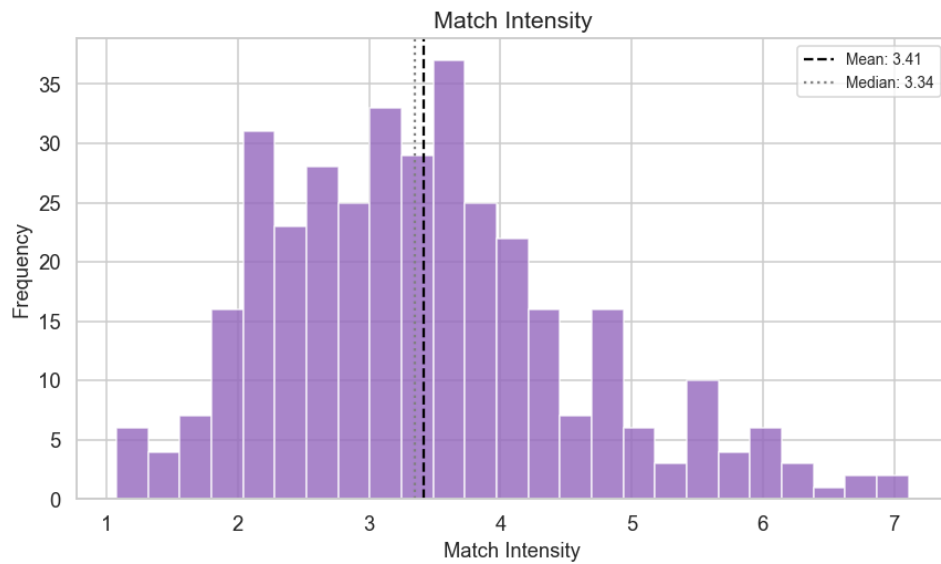
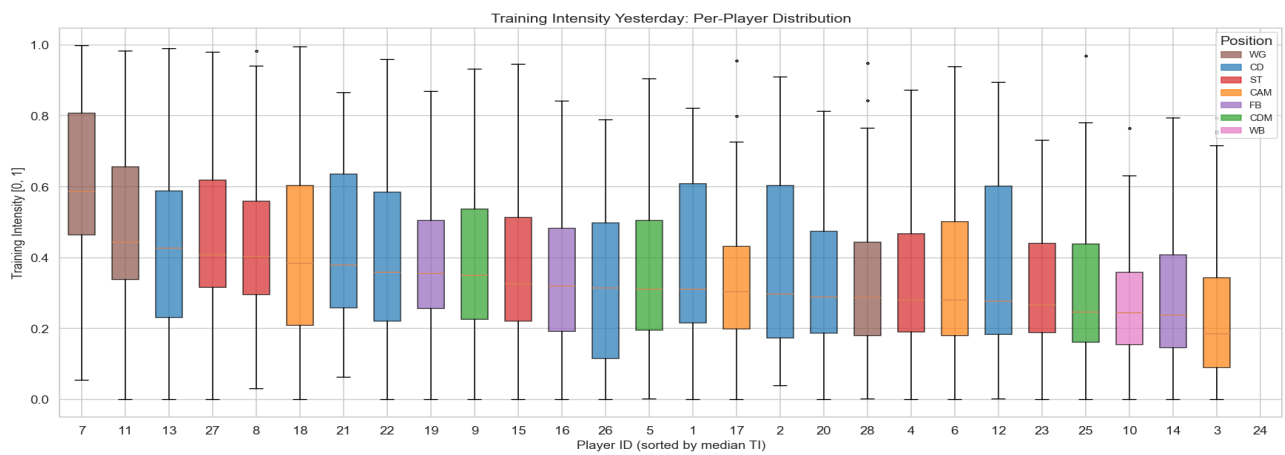


Figure 2: Training Intensity Yesterday per player, sorted by median. Coloured by playing position.



3. Experiment 1 — Match Intensity Prediction

Question: Can we predict how hard a player will perform in the next match, given their morning training state?

Why we tried this first: Before building a causal framework to optimise training load, we needed to verify whether the intended outcome — match-day performance — is even *predictable* from training and wellness data. If standard (non-causal) ML cannot predict it, then causal methods will certainly not identify a causal effect either. This experiment serves as that prerequisite check.

The notebook starts with a **diagnostic EDA** examining the prediction task: dataset scope, target distribution, feature-target correlations, and train/test distribution shift. This revealed that the effective dataset is small (~248 training, ~72 test) and features have near-zero correlation with the target.

Experiment A — Raw Match Intensity

We predicted Match Intensity Yesterday from morning covariates using Ridge Regression, XGBoost, CatBoost, and TabPFN across 10 lag values. Player ID was included as a feature.

Model	Best Lag	RMSE	Null RMSE	R2	Skill
TabPFN	8	1.1097	1.3820	0.272	+19.7%
CatBoost	5	1.1932	1.3820	0.158	+13.7%
LinReg	1	1.2250	1.3820	0.113	+11.4%
XGBoost	2	1.2412	1.3820	0.089	+10.2%

TabPFN achieves $R^2 = 0.27$ — seemingly useful. However, feature importance analysis reveals that **Player ID is the dominant feature**: the model mostly learns that some players consistently perform higher than others, regardless of training.

Experiment B — Personal Deviation

To test whether there is signal *beyond* player identity, we defined **Match Intensity Personal Deviation** = a player's match intensity minus their own expanding-mean baseline. This removes the player fixed effect: the model must now predict whether a player performs above or below *their own average*.

Result: all models achieve R^2 near zero or negative on the deviation target. With the player effect removed, there is **no recoverable signal** linking morning state to within-player match performance variation. Per-player median $R^2 = -0.75$; only 1 of 17 players has $R^2 > 0$.

Full results, permutation importance (LOO ablation, greedy forward selection), and per-player breakdowns are in **notebook 2.1. Experiment1**.

Conclusion

Match-day performance is NOT predictable from training and wellness data. Experiment A showed apparent signal ($R^2=0.27$) driven entirely by Player ID. Experiment B removed this effect and found no signal. Since the outcome is not predictable in a non-causal sense, it cannot serve as a causal target. This validates redirecting the framework toward proximal outcomes.

4. Experiment 2 — Training Intensity Prediction

Given the negative result of Experiment 1 — match-day performance is not predictable from training data — we pivoted to a more achievable and equally valuable question.

Question: Can we recover the coaching staff's implicit load-assignment policy from observable morning state? And if so, can the model's output serve as a **data-driven proxy for player readiness to train**?

Why this matters: Instead of trying to directly optimise match performance (which proved intractable), we can capture the coaching expertise embedded in 20 months of daily decisions. If morning covariates (wellness, fatigue, ACWR, schedule position) predict what the coaches actually prescribe, then: (1) the model **encodes the coaching staff's collective judgment** about player readiness, and (2) the predicted score becomes a continuous **Readiness to Train** metric — if the model predicts high intensity, the player is in a state where the coaches would have prescribed a hard session.

Setup: We predicted Training Intensity Yesterday (continuous, [0,1)) from 28 morning-assessment covariates using four models: Ridge Regression, XGBoost, CatBoost, and TabPFN. Each was tested across 3 lag values (1, 2, 3 days of history). The same processed RTT.xlsx dataset was used as in all other experiments, but the effective sample size is much larger than Experiment 1 because every training day contributes (not just match days).

Results

Results across all model-lag combinations show that the coaching policy **is substantially recoverable** from morning covariates. The best models (TabPFN, CatBoost) achieve R2 values in the range 0.35–0.45, meaning 35–45% of the variance in prescribed training intensity is explained by observable player state.

Key Finding

The coaching staff's load decisions are systematic and largely driven by observable physiological signals. The fitted model can serve as a continuous "Readiness to Train" score for each player-day. This is the primary actionable output of the project.

Feature importance analysis (permutation importance, 10 repeats on test set) reveals the **top drivers of coaching decisions**:

Rank	Feature	Importance	Interpretation
1	Days Until Match	0.0102	Strongest driver: coaches taper intensity as match approaches
2	Days Until Match (t-1)	0.0061	Previous day's schedule position also matters
3	Total Distance % Yesterday	0.0039	Yesterday's GPS load volume (% of match benchmark)
4	Total Distance (m) Yesterday	0.0021	Raw distance in metres from yesterday's session
5	Total Distance % Yesterday (t-1)	0.0015	Two-day load history contributes
6	High Speed Distance (m) Yesterday	0.0013	High-speed running volume from previous session
7	Total Minutes Yesterday	0.0011	Session duration as a load indicator
8	Activity Type Yesterday (t-1)	0.0009	Type of session two days ago (game, training, recovery)
9	Position (t-1)	0.0008	Playing position influences prescribed load
10	HSD % Yesterday (t-1)	0.0008	Two-day high-speed distance history

The key insight: **match-cycle position** (Days Until Match) is by far the strongest predictor, confirming that coaches follow a clear periodisation pattern. GPS load history (total distance, high-speed distance) ranks next, indicating that coaches also account for cumulative recent load when prescribing intensity.

Full results tables, SHAP/permutation importance plots, and per-player breakdowns are available in **notebook 2.2. Experiment2** with all visualisations preserved.

Practical use

The Experiment 2 model can be deployed as a **decision-support tool**: given a player's morning state, the model outputs a predicted training intensity that reflects what the coaching staff would typically prescribe. Deviations between the model's prediction and the actual prescription highlight cases where the coach is making an unusual decision — which may warrant a second look.

5. Experiment 3 — Status Decrease Prediction

As an auxiliary experiment, we investigated whether training load affects short-term player health outcomes. This experiment frames the problem explicitly in causal terms.

What is Status Decrease? A binary indicator (0/1) capturing whether a player's medical status worsened from one day to the next (e.g., Available → Attention, or Attention → Injured). This is a rare event: only about 3–5% of player-days show a status decrease.

Causal framing: In this experiment, **Training Intensity Yesterday** is explicitly treated as the **treatment variable (A)**, while morning wellness, ACWR, schedule position, and other player-state features serve as **covariates (L)**. Status Decrease is the **outcome (Y)**. The experiment uses a single lag (lag=1) to keep the model interpretable.

Two modes are run:

- **Prediction mode** — uses only morning covariates (L) to predict tomorrow's status (early-warning system)
- **Causal framing mode** — adds Training Intensity (A) as treatment variable alongside covariates (L) to examine whether yesterday's load explains today's status change

Results

Status Decrease is a challenging prediction target due to the severe class imbalance (~97% negative, ~3% positive). Models achieve ROC-AUC values above the 0.5 random baseline but with limited precision at the decision threshold. The causal framing mode reveals an important phenomenon: the coefficient on Training Intensity Yesterday is **negative** in the logistic regression — higher training intensity appears to *reduce* the risk of status decrease.

This counterintuitive result is a textbook example of **confounding by indication**: coaches prescribe higher intensity to players who are already in good condition. The raw association between load and outcome conflates the physiological effect of training with the coach's selection of who receives hard training. Extracting the true causal effect requires the propensity model from Experiment 2 combined with G-computation or IPTW methods.

Conclusion

The negative coefficient on Training Intensity demonstrates exactly why causal methods are needed: naive regression attributes the coach's good judgment (prescribing hard sessions to healthy players) to the training itself. Extracting the true causal effect of training intensity on player health requires the propensity model from Experiment 2 combined with G-computation or IPTW methods. Full results, ROC/PR curves, and per-player breakdowns are in notebook 2.3. Experiment3.

6. Conclusions and Recommendations

What we achieved

- **Data integration:** Merged four separate monitoring datasets into a unified, analysis-ready file with 46 engineered features per player-day, fully reproducible from raw data.
- **Experiment 1** confirmed that match-day performance is not predictable from training data alone ($R^2 = 0.27$, driven by player identity). This ruled out Match Intensity as a causal target and validated the focus on proximal outcomes.
- **Experiment 2** successfully recovered the coaching staff's load-assignment policy from morning covariates ($R^2 = 0.35\text{--}0.45$). The model can serve as a continuous "Readiness to Train" score and as a propensity model for causal estimation. Feature importance analysis confirmed that periodisation structure (days until match) and workload ratios (ACWR) are the primary drivers of coaching decisions.
- **Experiment 3** built an outcome model for next-day Status Decrease and demonstrated the presence of confounding by indication in the raw data — confirming that causal methods are necessary to isolate the true effect of training load.

Recommendations for OH Leuven

1. **Deploy the Experiment 2 model as a decision-support tool.** The predicted Training Intensity score from Experiment 2 provides a data-driven "expected load" for each player-day. When the actual prescribed load deviates significantly from the prediction, this flags an unusual coaching decision that may warrant review.
2. **Use the Status Decrease model as an early-warning system.** Although the prediction is imperfect (due to low base rate), flagging players with elevated risk before the training session adds a safety layer to the monitoring process.
3. **Continue data collection.** More seasons of data will improve model stability and enable validation across different squad compositions and fixture schedules.
4. **Next research step: causal estimation.** Combining the propensity model (Experiment 2) with the outcome model (Experiment 3) via G-computation or Inverse Probability of Treatment Weighting (IPTW) will allow estimation of the true causal effect of training intensity on player outcomes — free from the confounding by indication demonstrated in Experiment 3.